

Written HW5

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I use Neovim (BTW)

Full HW repository at
github.com/LoganCTanner/School

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1. Suppose X and Y have the joint density function $f_{X,Y}(x,y) = e^{-(x+y)}$ for $x \geq 0, y \geq 0$. Let $W = X + Y$.

(a) Are X and Y independent? Justify.

Answer:

If $f_X(x) \cdot f_Y(y) = f_{X,Y}(x,y)$, then they are independent.

$$\begin{aligned}f_X(x) &= e^{-x} \int_0^{\infty} e^{-y} dy \\&= e^{-x} (-e^{-y}|_0^{\infty}) \\&= e^{-x} (0 - (-1)) \\&= e^{-x}\end{aligned}$$

$$\begin{aligned}f_Y(y) &= e^{-y} \int_0^{\infty} e^{-x} dx \\&= e^{-y} (-e^{-x}|_0^{\infty}) \\&= e^{-y} (0 - (-1)) \\&= e^{-y}\end{aligned}$$

$$\begin{aligned}f_X(x) \cdot f_Y(y) &= e^{-x} \cdot e^{-y} \\&= e^{-(x+y)} \\&= f_{X,Y}(x,y)\end{aligned}$$

$\therefore f_{X,Y}(x,y)$ is independent.

(b) Find the cdf of W .

Answer:

$$\begin{aligned} F_W(w) &= \int_0^w \int_0^{w-x} e^{-x-y} dy dx \\ &= \int_0^w e^{-x} (-e^{-y}|_0^{w-x}) dx \\ &= \int_0^w e^{-x} (1 - e^{x-w}) dx \\ &= \int_0^w e^{-x} - e^{-x+x-w} dx \\ &= \int_0^w e^{-x} - e^{-w} dx \\ &= -e^{-x} - xe^{-w}|_0^w \\ &= -e^{-w} - we^{-w} - (-1 - 0) \\ &= -e^{-w} - we^{-w} + 1 \end{aligned}$$

(c) Use (b) to find the pdf of W .

Answer:

$$\begin{aligned} f_W(w) &= -e^{-w} - we^{-w} + 1dx \\ &= e^{-w} - (e^{-w} + (-we^{-w})) \\ &= e^{-w} + we^{-w} - e^{-w} \\ &= we^{-w} \end{aligned}$$

(d) Now find the pdf of W using the convolution formula (from the beginning of Chapter 6.3).

2. A bin of five transistors is known to contain two that are defective. They are to be tested, one at a time, until the defective ones are identified. Let X be the number of tests needed to find one defective transistor, and let Y be the additional tests needed to find the second. Note that it's possible that a defective transistor can be found even if it wasn't tested, by process of elimination.

(a) Find the joint pmf of X and Y .

Answer:

Letting the probability of X being the first faulty transistor, then

$$\begin{aligned} P(X = 1) &= \frac{2}{5} \\ P(X = 2|(X = 1)^C) &= \frac{2}{4} \cdot \frac{3}{5} = \frac{6}{20} \\ P(X = 3|(X = 2)^C \cap (X = 1)^C) &= \frac{2}{3} \cdot \frac{3}{5} \cdot \frac{2}{4} = \frac{12}{60} \end{aligned}$$

Should the first 3 transistors work the remaining 2 must be faulty. Similarly for Y being the number of tests to find the second faulty transistor,

$$\begin{aligned} P(Y = 1|X = 1) &= \frac{1}{4} \div \frac{2}{5} = \frac{5}{8} \\ P(Y = 1|X = 2 \cap (Y = 1)^C) &= \frac{1}{3} \div \frac{2}{4} = \frac{4}{6} \\ P(Y = 1|X = 3) &= \frac{1}{2} \div \frac{2}{3} = \frac{3}{4} \\ P(Y = 1) &= \frac{5}{8} \cdot \frac{2}{5} + \frac{4}{6} \cdot \frac{2}{4} + \frac{3}{4} \cdot \frac{2}{3} \\ &= \frac{10}{40} + \frac{7}{30} + \frac{6}{12} \\ &= \frac{30 + 28 + 60}{120} = \frac{118}{120} \end{aligned}$$

It follows for $Y = 2$ that

$$\begin{aligned} P(Y = 2|X = 1) &= \frac{1}{3} \cdot \frac{2}{5} = \frac{5}{6} \\ P(Y = 2|X = 2) &= \frac{1}{2} \div \frac{2}{4} = 1 \\ P(Y = 2) &= \frac{5}{6} \cdot \frac{2}{5} + \frac{1}{1} \cdot \frac{2}{4} \\ &= \frac{r}{15} + \frac{1}{2} \\ &= \frac{8}{60} + \frac{15}{60} \\ &= \frac{23}{60} \end{aligned}$$

With $P(Y = 2|X = 3) = 1$ given by our deduction, that is to say that if transistor 1 works, transistor 2 is faulty, and transistor 3 is good. Then either transistors 4 or 5 must be faulty, and if transistor 4 works then 5 must be faulty. $P(Y = 3|X = i \in \{1, 2, 3\})$ would

then be

$$\begin{aligned}P(Y = 3|X = 1) &= \frac{1}{2} \\P(Y = 3) &= \frac{1}{2} \cdot \frac{2}{5} \\&= \frac{1}{5} = \frac{12}{60}\end{aligned}$$

With the undefined $P(Y = 3|X = 3)$ not being within the scope of the problem with only 5 transistors, and a similar deductive process for $P(Y = 3|X = 2) = 1$.

3. Let X and Y be independent *uniform* $[0, 1]$ random variables.

(a) Find pdf of $|X - Y|$.

Answer:

With the marginal pdfs of X and Y both being $f = \frac{1}{1-0} = 1$, and their same intervals over $[0, 1]$, it follows that the cdf of $|X - Y|$ would then be

$$\begin{aligned}
 P(|X - Y| \leq a) &= P(-a \leq X - Y \leq a) \\
 &= 2P(Y \leq X + a) \\
 &= 2 \int_0^{1-a} \int_{x+a}^1 dy dx \\
 &= 2 \int_0^{1-a} 1 - x - a dx \\
 &= 2 \left(x - \frac{1}{2}x^2 - ax \Big|_0^{1-a} \right) \\
 &= 2 \left(x(1-a) - \frac{1}{2}x^2 \Big|_0^{1-a} \right) \\
 &= 2 \left((1-a)^2 - \frac{1}{2}(1-a)^2 \right) \\
 &= 2 \left(1 - \frac{1}{2} \right) (1-a)^2 \\
 &= 1 - 2a + a^2
 \end{aligned}$$

and the pdf is then obtained from the derivative of the cdf

$$\begin{aligned}
 f_{|X-Y|}(a) &= (1 - 2a + a^2) da \\
 &= \boxed{2 - 2a}
 \end{aligned}$$

(b) Compute $E[|X - Y|]$.

Answer:

Given the pdf of $2 - 2a$, the expected value would then be

$$\begin{aligned}
 E[f_{|X-Y|}(a)] &= \int_0^1 a(2 - 2a) da \\
 &= a^2 - \frac{2}{3}a^3 \Big|_0^1 \\
 &= 1 - \frac{2}{3} = \boxed{\frac{1}{3}}
 \end{aligned}$$